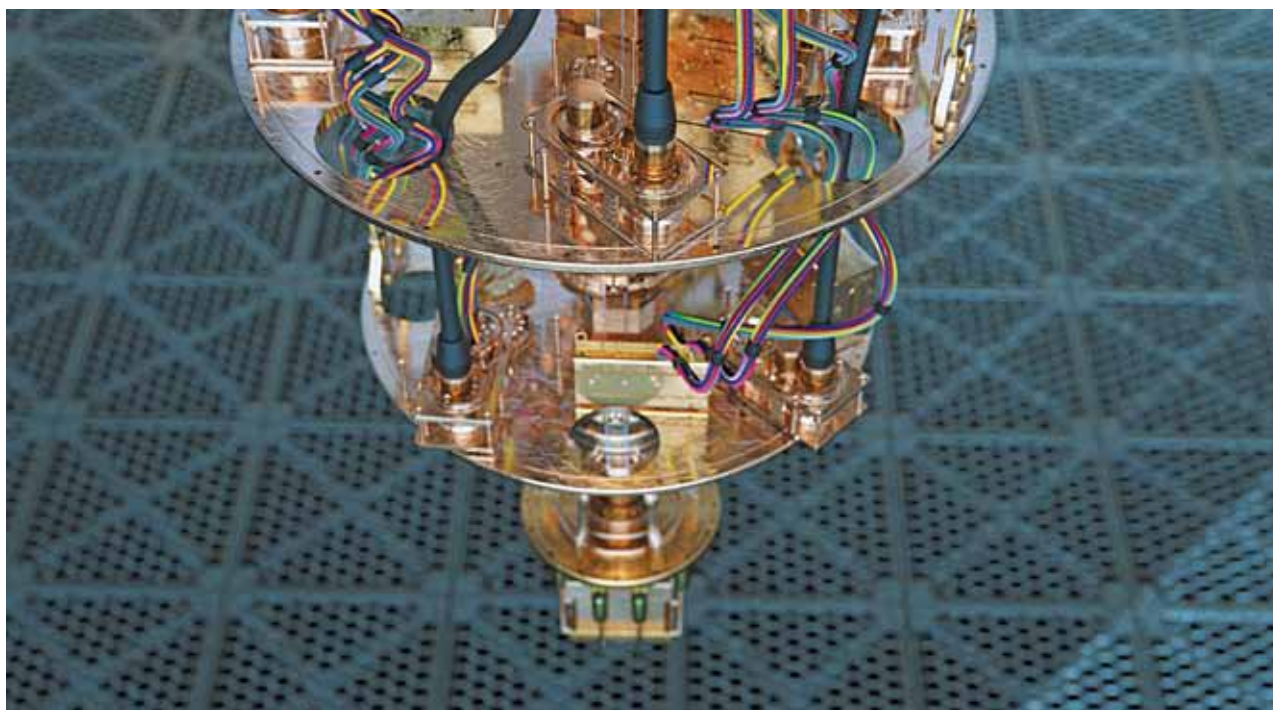


New Approaches to Building Quantum Computers

There are lots of ways to construct qubits. Collaboration between academia and industry is going to play a major role in how this shapes out. Depending on each of their strengths, academic labs can focus on supplying the core expertise necessary to get a quantum computer going while the industry can provide the engineering muscle to build such an intricate machine.



Qubits are at the heart of quantum computers that use the properties of quantum physics to store data and perform computations. These computers are believed to be able to solve certain critical computational problems substantially faster than the classical computers in use. Hence there is a huge interest in them and many countries are investing in building them.

This article updates the list of global players in the field and describes some of the new technologies that are being developed. So far, the small prototypes of quantum computers that have been

developed by academia and industry in the USA and some other countries have affirmed the proof-of-concept for investing in their further development.

Advancements made by some of the players in the field are summarised below.

1. The Canadian firm **D-Wave** has announced that it is commercialising its next-generation system, Advantage, built for businesses. The company's computer is already being used, via its cloud, by some corporations to build and run applications for their operations. Now D-Wave is making Advantage available to other businesses as well. The system is made of more than 5000 qubits, which is more than double the count of processing units in its earlier 2000Q machine. Each of these qubits will connect with 15 others, increasing the connectivity by two-and-a-half times its predecessor's, leading to larger, denser, and more powerful graphs for building commercial applications.
2. **IONQ**, a firm in Maryland, USA, uses trapped ytterbium ions manipulated by lasers to perform the calculations.